



Organization of Gene

“Fundamental Biochemistry of the Genome”

SGL - 2

Hawler Medical University
College of Medicine

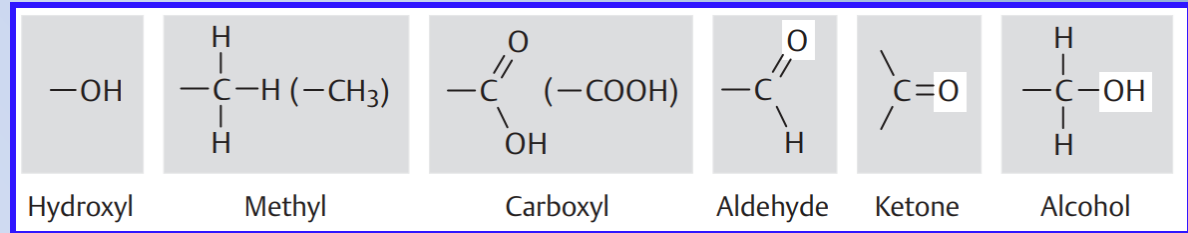
Dr. Bejan Ahmed DIZAYEE
M.Sc. Medical Microbiology / Molecular Biology
Director of Central Veterinary Laboratory in Kurdistan Region

Some Types of Chemical Bonds

A. Compounds of (H), (O), & (C)

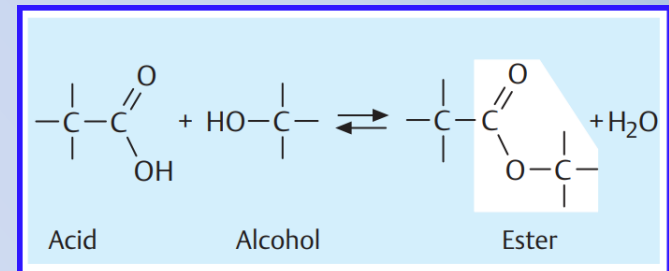
Four simple combinations of these atoms occur frequently in biologically important molecules:

They impart to the molecules characteristic chemical properties, including possibilities to form compounds.



B. Acids and esters

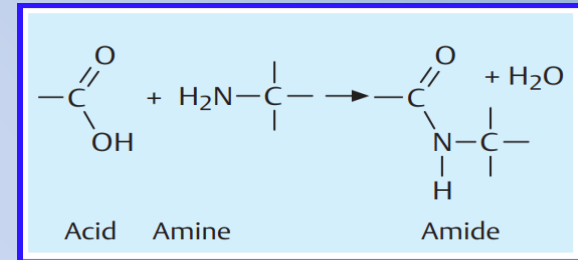
Many biological substances contain a carbon–oxygen bond with weak acidic or basic (alkaline) properties. The degree of acidity is expressed by the pH value, ester is formed when an acid reacts with an alcohol. Esters are frequently found in lipids and phosphate compounds.



Some Types of Chemical Bonds

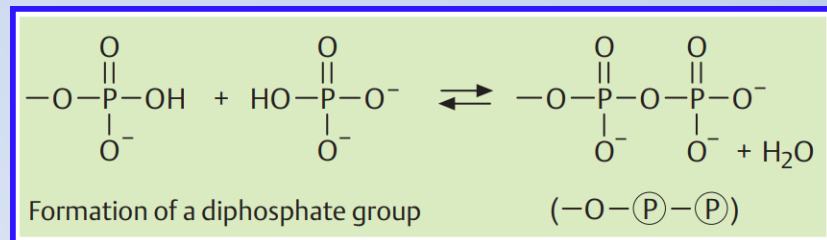
C. Carbon–nitrogen bonds (C—N)

C—N bonds occur in amino groups, amines, and amides, especially in proteins. Of paramount significance are the amino acids, which are the subunits of proteins. All proteins have a specific role in the functioning of an organism.



D. Phosphate compounds

Ionized phosphate compounds play an essential biological role. HPO_4^{2-} is a stable inorganic phosphate ion from ionized phosphoric acid. A phosphate ion and a free hydroxyl group can form a phosphate ester. Phosphate compounds play an important role in energy-rich molecules because they can store energy



DNA & RNA Composition :

Nucleotides: “Overview”

Nucleotides participate in almost all biological processes. They are:

1. Convey energy (e.g. ATP, GTP).
2. Encoding, transmitting and expressing genetic information
3. They are part of essential coenzymes,
4. They regulate numerous metabolic functions (enzymes, RNA).

Since all these functions are based on genetic information of the cells, nucleotides represent a central class of molecules for genetics.

Nucleotides are composed of three integral parts:

- 1- Sugars (Five-C-atom sugar residue ***Ribose*** or ***Deoxyribose***).
- 2- ***Phosphates*** Groupe.
- 3- Nucleotide base (***Purine*** or ***Pyrimidine***) bases.

Molecular Biology : Fundamentals



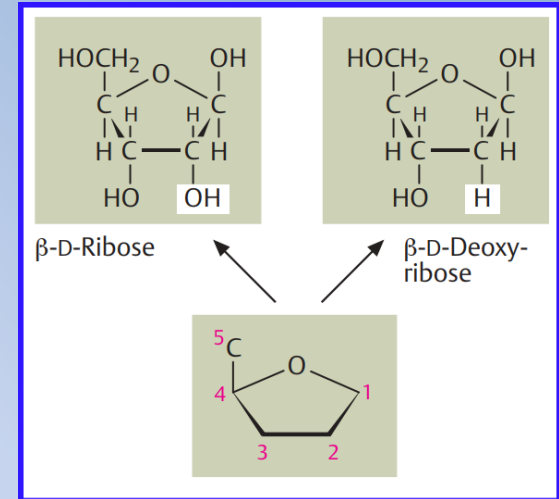
DNA & RNA Composition : Nucleotides

The Nucleotides compose of:

A. Sugar residues

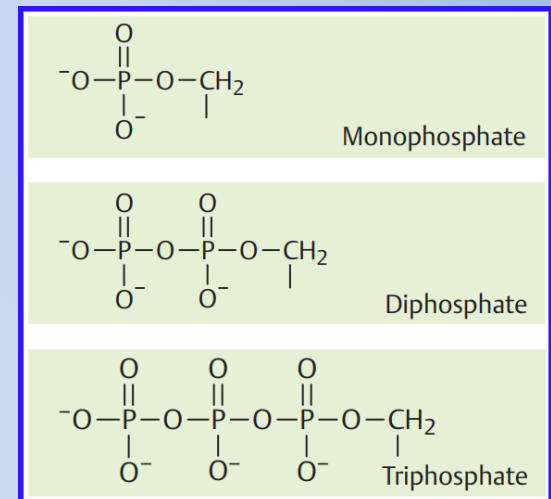
The sugar residues in nucleotides are usually derived from either

- Ribose (in ribonucleic acid, RNA) or
- Deoxyribose (in deoxyribonucleic acid, DNA).



B. Phosphate groups

Phosphate groups may occur alone (monophosphates), in twos (diphosphates) or in threes (triphosphates). They are normally bound to the hydroxy group of the C atom in position 5 of a five-C-atom sugar (pentose).

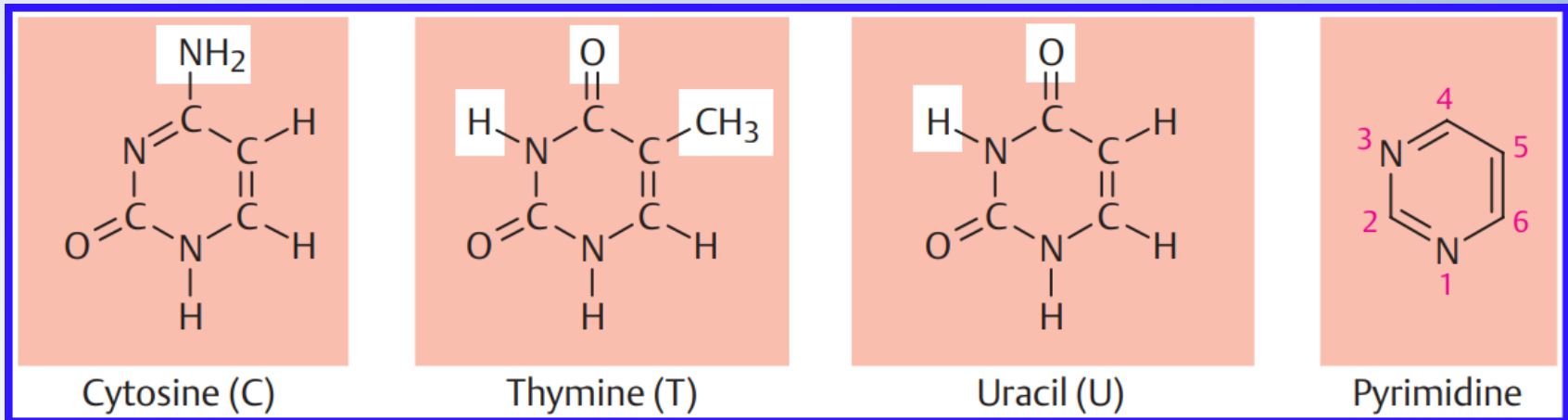


DNA & RNA Composition : Nucleotides

C. Nucleotide bases of pyrimidine

The Nitrogen base in nucleotides are usually cytosine (C), thymine (T), and uracil (U) are the three pyrimidine nucleotide bases.

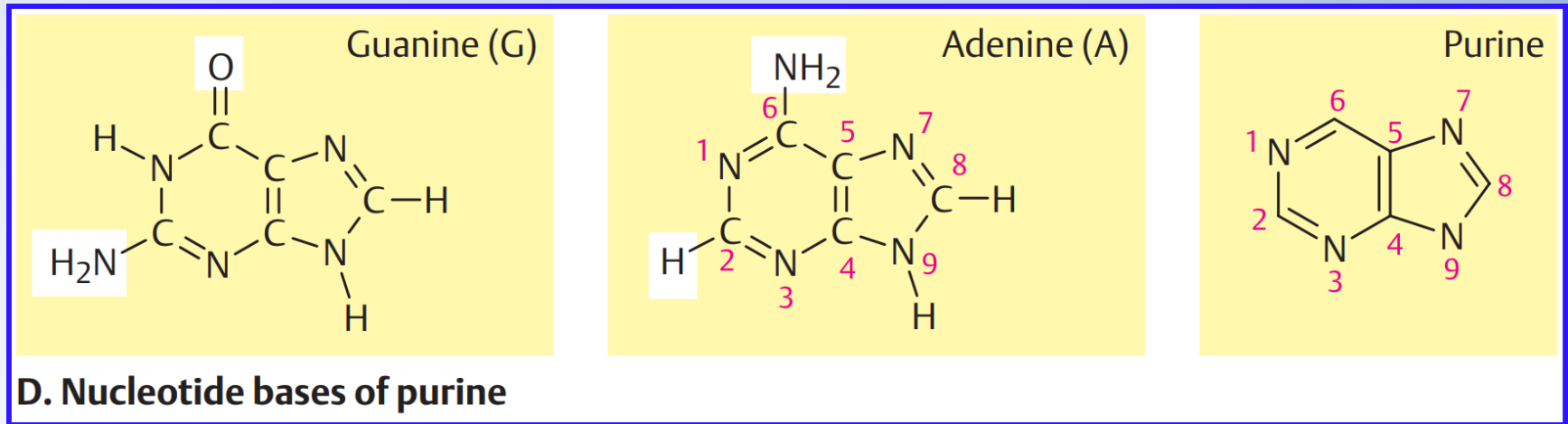
(in uracil) and in the presence or absence of a double bond between N3 and C4 (present in cytosine)



DNA & RNA Composition : Nucleotides

D. Nucleotide bases of purine

Adenine (A) and guanine (G) are the two nucleotide bases of purine. They differ in their side chains and a double bond (between N1 and C6)



DNA & RNA Composition: Nucleosides

Nucleosides

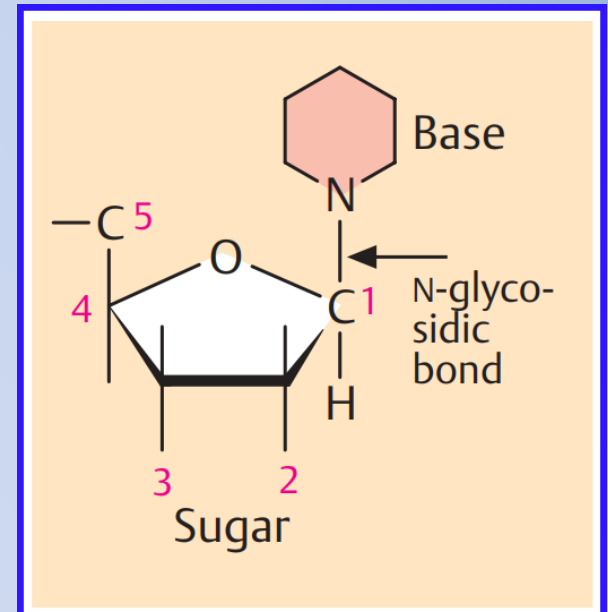
A nucleoside is a compound of a sugar residue (ribose or deoxyribose) and a nucleotide base.

The bond is between the C atom in position 1 of the sugar (as in compounds of sugars) and an atom of the base (N-glycosidic bond).

The nucleotides of the various bases are named according to:

- Whether they are a ribonucleoside or a deoxyribonucleoside, e.g.,
- Adenosine or Deoxyadenosine,
 - Guanosine or Deoxyguanosine,
 - Uridine (occurs only as a Ribonucleoside),
 - Cytidine or Deoxycytidine.
 - Thymidine occurs only as a Deoxynucleoside.

base + sugar



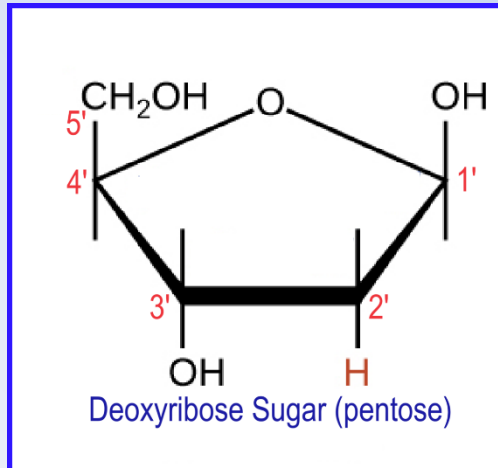
Molecular Biology : Fundamentals

DNA & RNA Composition: Nucleotides

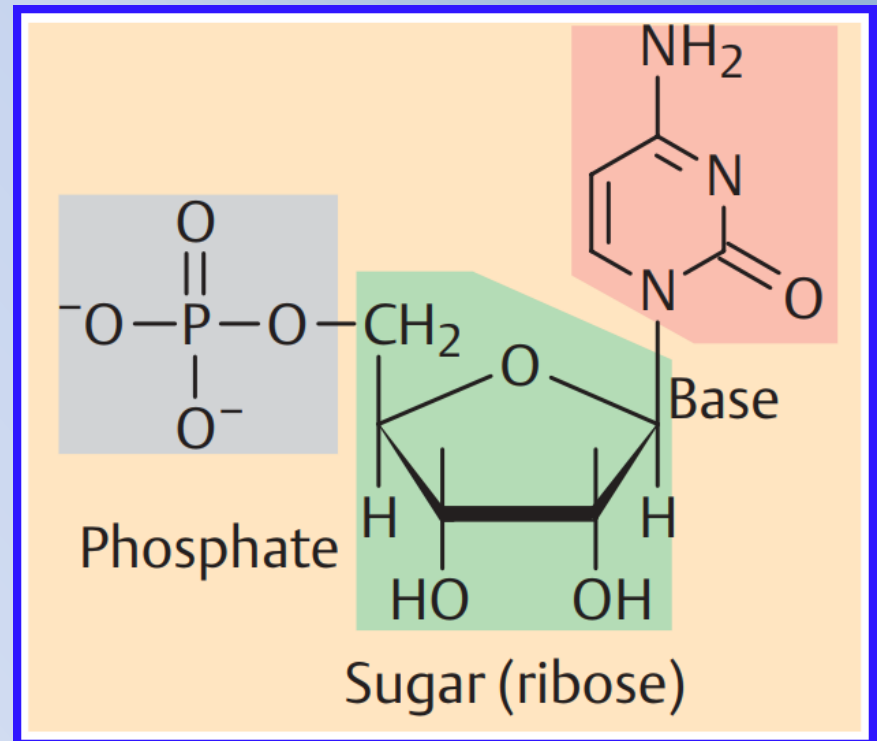


Nucleotide

A nucleotide is a compound of a five-C-atom sugar residue (ribose or deoxyribose) attached to a nucleotide base (pyrimidine or purine base) and a phosphate group. Nucleotides are the subunits of nucleic acids.



Sugar residues (pentose)

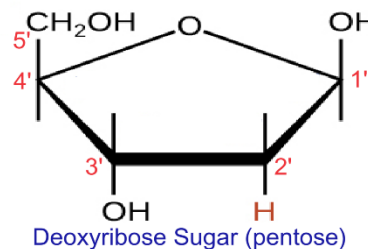
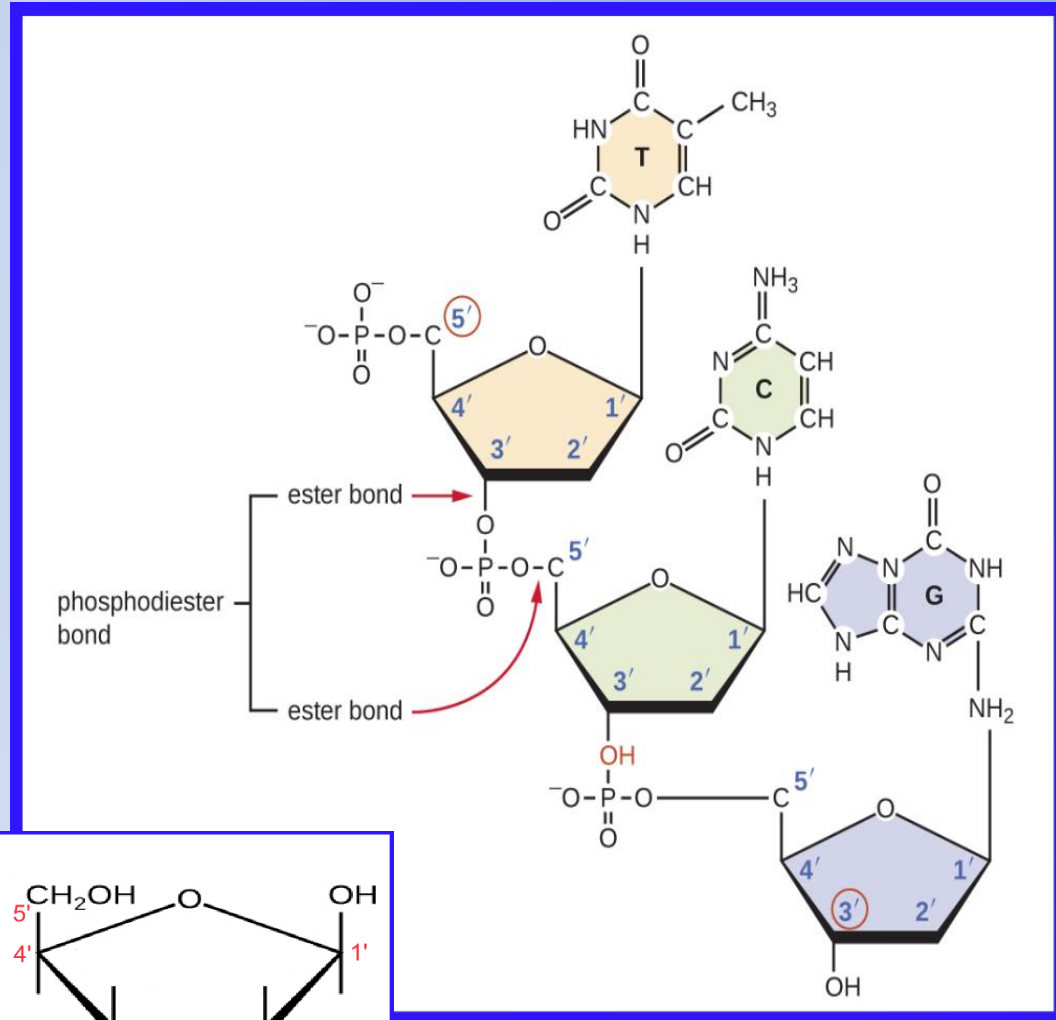


N-base + sugar + phosphate

DNA & RNA Composition: Nucleic acids

Nucleic acids are formed when nucleotides are joined to each other by means of phosphodiester bridges between the 3' C atom of one nucleotide and the 5' C atom of the next.

The linear sequence of nucleotides is usually given in the 5' to 3' direction with the abbreviations of the respective nucleotide bases. For instance, ATCG would signify the sequence adenine (A), thymine (T), cytosine (C), and guanine (G) in the 5' to 3' direction.



DNA Feature:

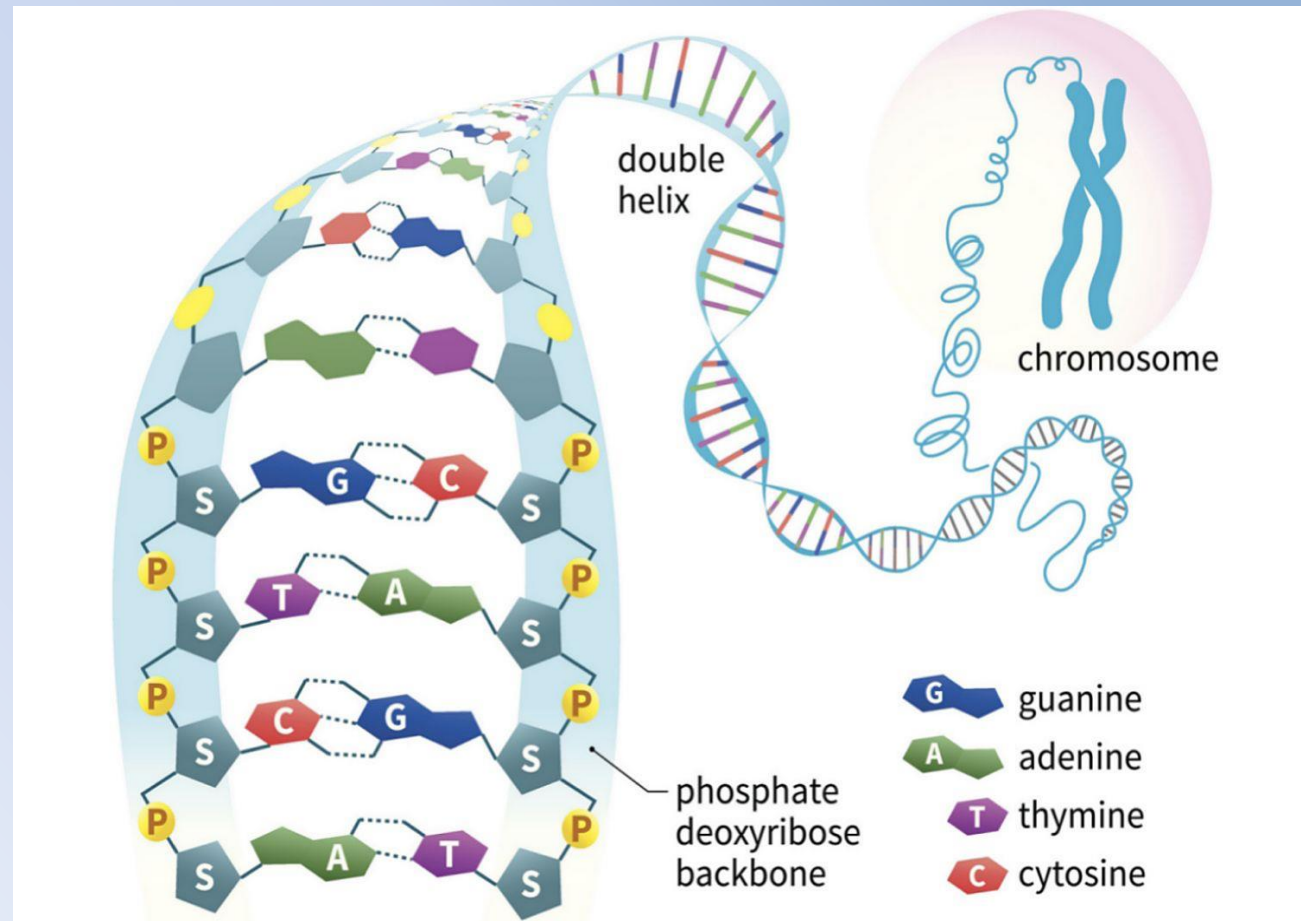
Deoxyribo Nucleic Acid (DNA) is a complex long-chained double helix molecule,
Its chemical composition:

A- Nucleotide

B- Nucleotide chain

C- Special Relationship

D- DNA double strand
Back Bone

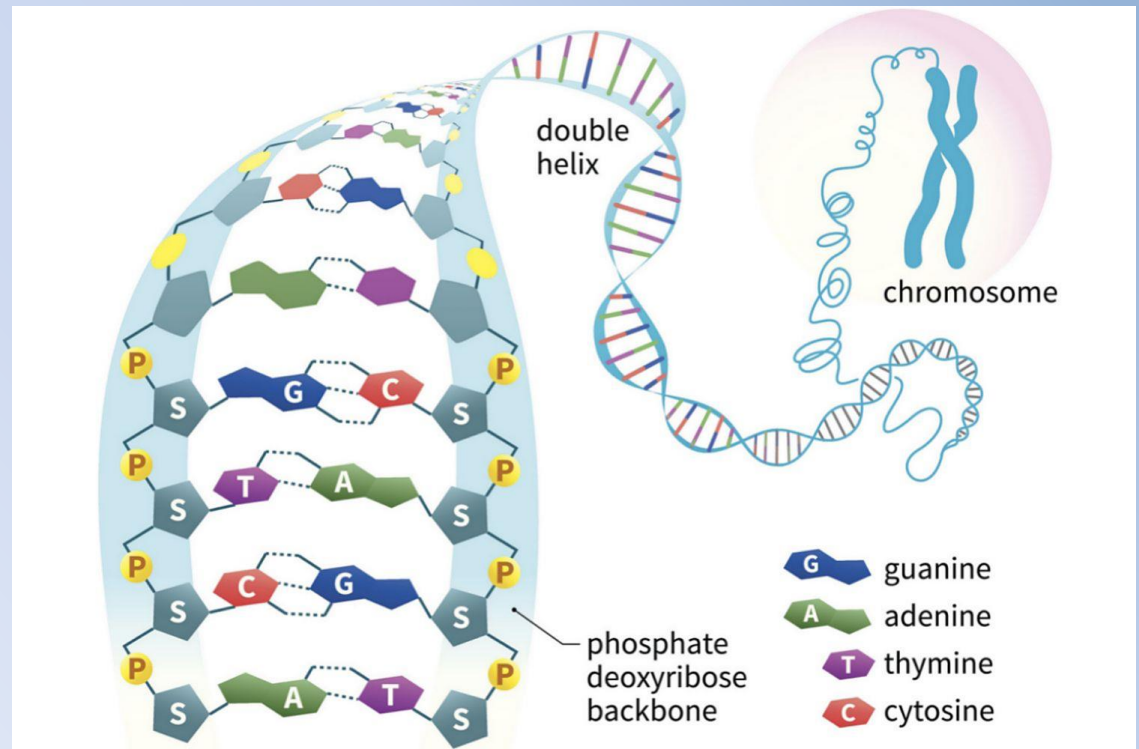


A- Nucleotide

A nucleotide is a compound of a five-C-atom sugar residue (ribose or deoxyribose) attached to a nucleotide base (pyrimidine or purine base) and a phosphate group. Nucleotides are the subunits of nucleic acids.

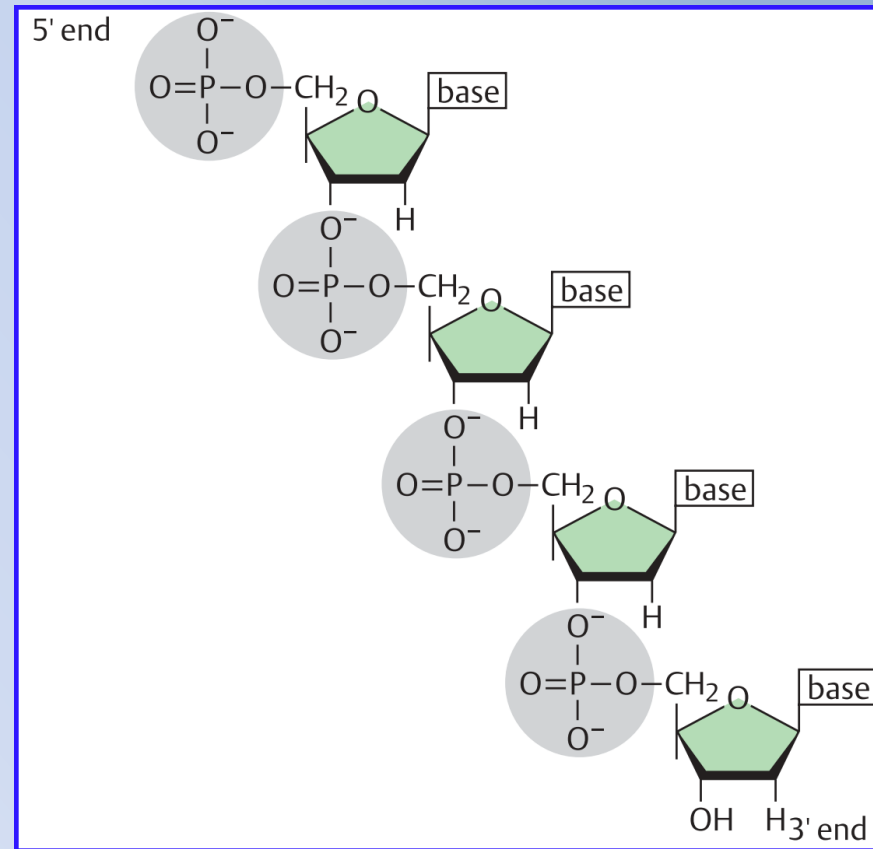
- Nucleotide bases
- Sugar (deoxyribose)
- Phosphate groups

are determine the three-dimensional structure of DNA, from which it derives its functional consequence



B. Nucleotide chain

- DNA is a polymer of deoxyribonucleotide units.
- The nucleotide chain is formed by joining a hydroxyl group on the sugar of one nucleotide to the phosphate group attached to the sugar of the next nucleotide.
- The variable part is in the sequence of the nucleotide bases A, T, C, and G.
- The phosphate group at position C5 (the 5' carbon) of one sugar joins to the hydroxyl group at position C3 (the 3' carbon) of the next sugar by means of a phosphate diester bridge. the sequence of nucleotide bases is written in the 5' to 3' direction.



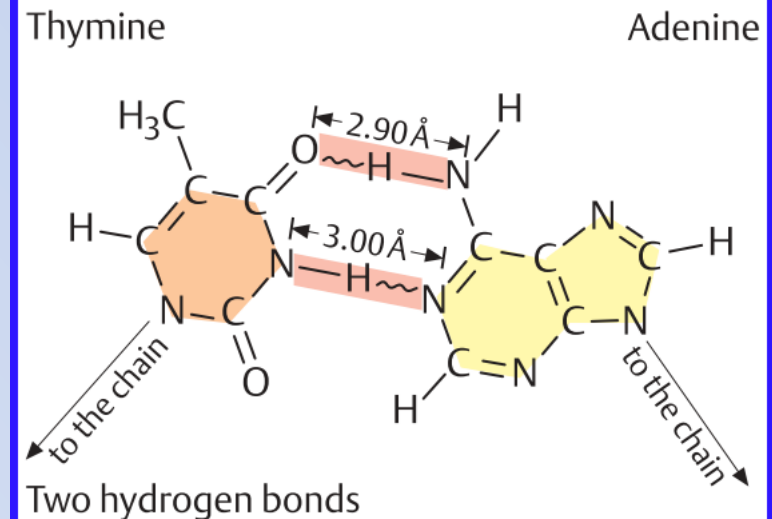
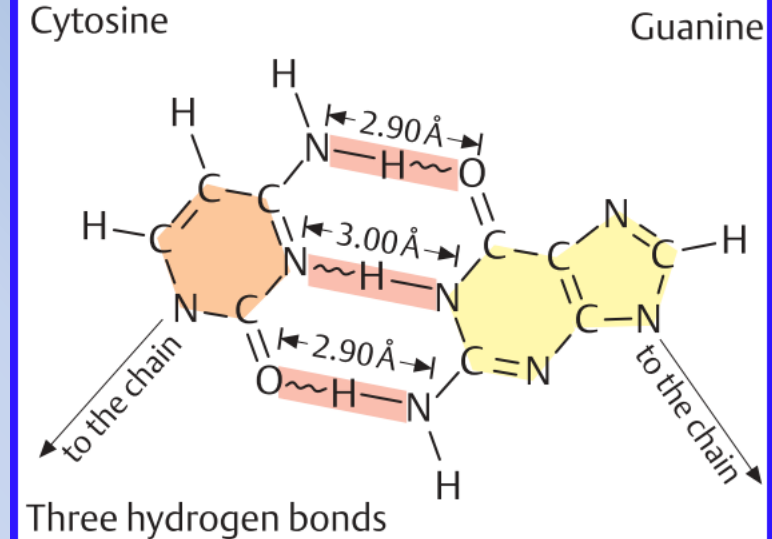
Molecular Biology : Fundamentals

DNA Structure & Feature:

C. Special Relationship

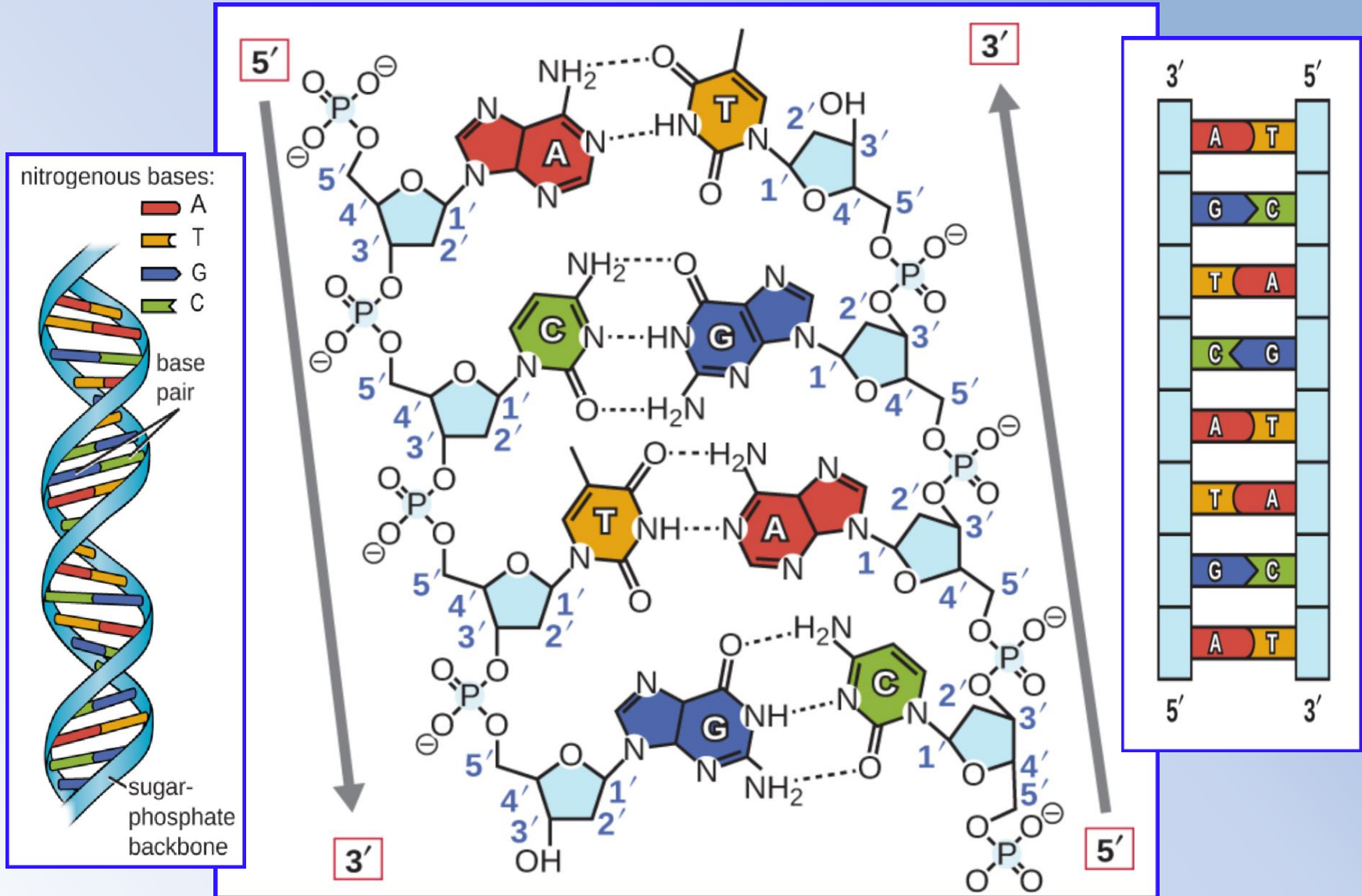
DNA double strand & (base pairs "bp")

- DNA forms a double strand. As a result of the spatial relationships of the nucleotide bases.
- A regular double helix requires a purine in one strand to be matched by a pyrimidine in the other, so the purine (A or G) always lies opposite a pyrimidine (T or C).
- Three hydrogen-bond bridges are formed between C and G, and two between T and A .
- Therefore, they pair with each other in a (complementary base pairs G–C and A–T). Other spatial relationships are not usually possible.
- The specificity of base pairing is the most important structural characteristic of DNA.



Molecular Biology : Fundamentals

DNA Structure & Feature: C. Special Relationship



D. DNA double strand Back Bone

- DNA is a very stable molecule. Scientists routinely send DNA samples in the post without worrying about refrigeration
- DNA of high enough quality to be cloned has been recovered from frozen mammoths and mummified Pharaohs thousands of years old
- This stability is provided by the robust repetitive phosphate±sugar backbone in each DNA strand, in which the phosphate links the 5' position of one sugar to the 3' position of the next

